

Boston House Price Prediction Using Linear Regression



Introduction & Problem Statement

This project predicts the **median house price (MEDV)** using the Boston Housing Dataset and Linear Regression, leveraging different housing features to build an accurate predictive model.

Dataset Features

CRIM Crime rate	ZN Residential land zone
INDUS Industrial area proportion	CHAS Charles River dummy variable
NOX Nitric oxide concentration	RM Average rooms
AGE Age of houses	DIS Distance to employment centers
RAD Highway accessibility	TAX Property tax rate
PTRATIO Student-teacher ratio	B Population statistic
LSTAT Lower status population %	MEDV Median house value

Technologies Used

 Python	 Pandas
 NumPy	 Matplotlib
 Seaborn	 Scikit-learn

Workflow

1	Load Dataset
2	Data Cleaning
3	EDA
4	Train-Test Split
5	Model Training
6	Prediction
7	Evaluation
8	Visualization

Data Preprocessing & EDA

Missing values are handled using **mean imputation**.
Exploratory Data Analysis includes:

- Correlation Heatmap
- Histograms
- Scatterplots
- Boxplots

Linear Regression Formula

Linear Regression predicts continuous values using:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Model Evaluation Metrics

MAE Mean Absolute Error — average magnitude of errors	MSE Mean Squared Error — penalizes larger errors more
RMSE Root Mean Squared Error — interpretable scale	R² Score Coefficient of Determination — model fit quality

✔ **Conclusion:** The project successfully predicts house prices using Linear Regression, demonstrating a complete end-to-end machine learning pipeline from data loading to evaluation and visualization.